**Sultan Qaboos University**
**College of Engineering**
Office of Assistant Dean for
Industrial Training and Community Services

**Weekly Report**
**Week #:** 6
Date:

| | | |
|---|---|---|
| Student | Full Name: Malek Al shukily | ID: 141776 |
| | Department: ☐CAE ☐ECE ☐MIE ☐PCE ■MCE | Mobile: 72471416 |
| Supervisor | Full Name: Younis Ghashim Al-Aghbari | Email: |
| | Training Company & Department: | Tel./Mobile: 24333710 |

**Tasks Completed (Briefly Describe any tasks that you completed during this week):**

- Home air conditioning components. .
- understanding maintenance procedure.
- learned expansion valve system.

**Tasks in Progress (if any):**

- —
-
-
- —
-
-

**Plan for next week (if different from this week):**

- Electronics and precision intruments.
- Electronic symbols.
- Passive & active components. .
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**Problems/Challenges/Recommendations (if any):**

No major problems we phase

**Supervisor Comments (your feedback is very important for our continuing improvement):**

No comments

**Student:**
I have read and understood the whole content of the Students' Training Manual.

Signature: ..............................

**Supervisor:**
I will do my best to guide the student towards achieving all required deliverables.

Signature: ..............................

**Note:** A copy of this report should be faxed to 24413416 or scanned and emailed to your SQU supervisor on a weekly basis. Originals should be placed in the Appendix of the **Final Report.**

# WEEKLY OPERATIONS & MAINTENANCE REPORT

Residential Air Conditioning Department — Technical & Field Services

| | | | |
|---|---|---|---|
| **Report Period:** | August 07 – August 13, 2026 | **Department:** | Residential HVAC Operations |
| **Prepared By:** | Lead HVAC Supervisor | **Target Units:** | Split, Window & Mini-VRF Systems |

| 142 | 38 | 98.5% | 4.9 / 5 |
|---|---|---|---|
| UNITS SERVICED | GAS TOP-UPS | FIRST-TIME FIX RATE | CUSTOMER RATING |

## 1. Department Overview & Executive Summary

During this operational week, the Residential Air Conditioning Department successfully completed maintenance and repair tasks across **142 residential AC units** (including split systems, window units, and residential multi-split VRF configurations). Operational efficiency remained high with a 98.5% first-time resolution rate.

Key focus areas this week included routine preventive maintenance, deep coil cleaning, electrical diagnostic checks, and gas refill procedures following leak detection and system evacuation. All technical operations strictly adhered to safe refrigerant handling protocols.

## 2. AC Maintenance Guidelines & Best Practices

Proper maintenance ensures optimal cooling efficiency, lowers energy consumption, and extends compressor lifespan. Field technicians follow a standardized 5-stage maintenance workflow:

1. **Initial Operational & Electrical Diagnostic:** Check current draw (amperage), line voltage, thermostat responsiveness, indoor fan motor speed, and initial supply/return air delta temperature ($\Delta T \ge 8^\circ\text{C}$ to $10^\circ\text{C}$).
2. **Air Filter & Blower Assembly Cleaning:** Remove, wash, and sanitize washable air filters. Inspect and clean the indoor centrifugal blower wheel to restore nominal airflow and prevent mold accumulation.
3. **Evaporator & Condenser Coil Deep Wash:** Apply non-acidic foam coil cleaner to evaporator and condenser fins. Wash thoroughly with low-pressure water to avoid fin damage, removing dust build-up and improving heat exchange efficiency.
4. **Condensate Drain Line Flushing:** Flush drain pan and drain pipe using high-pressure air or water. Clear algae or silt buildup and place anti-bacterial drain tablets to prevent future clogging and water leaks.
5. **Electrical & Mechanical Tightening:** Inspect start/run capacitors, contactors, wiring terminals, and fan motor bearings. Tighten loose connections and replace degraded capacitors exhibiting $> 5\%$ microfarad ($\mu\text{F}$) drop.

## 3. Refrigerant (Gas) Refilling Standard Operating Procedure (SOP)

Refrigerant gas top-up or complete recharge must only be performed after identifying and repairing gas leaks. Simply adding gas to a leaking system is strictly non-compliant with department standards.

### Step-by-Step Refilling Procedure:

1. **Leak Detection & System Isolation:** Pressurize the system with dry Nitrogen gas (up to 150–250 PSI depending on system specifications). Use bubble spray or electronic leak detectors to pinpoint leaks at flare joints, service valves, or coil bends. Repair leaks via brazing or flare re-seating.
2. **Deep System Evacuation (Vacuuming):** Connect dual-stage vacuum pump and digital micron gauge to the service ports. Pull a deep vacuum down to below **500 Microns** and hold for 15 minutes to eliminate all air and moisture from refrigerant lines.
3. **Refrigerant Type & Selection:** Identify exact refrigerant type from the unit nameplate (e.g., **R-410A**, **R-32**, or **R-22**).
    - *R-410A / R-32:* Synthetic blend / flammable blend — **must always be charged in liquid phase** (invert cylinder if non-dip tube).
    - *R-22:* Single component refrigerant — charged in vapor phase during system operation.
4. **Precision Weigh-In Charging:** Place refrigerant cylinder on a digital charging scale. Zero the scale and charge the exact weight (in grams/kg) specified on the manufacturer badge.
5. **Operating Pressure & Performance Verification:**
    - **R-410A Operating Suction Pressure:** $120 ext{--}140 ext{ PSI}$
    - **R-32 Operating Suction Pressure:** $115 ext{--}135 ext{ PSI}$
    - **R-22 Operating Suction Pressure:** $65 ext{--}75 ext{ PSI}$

    Verify superheat ($5 ext{--}8^\circ ext{C}$ for fixed orifice) or subcooling ($5 ext{--}7^\circ ext{C}$ for TXV) along with supply air temperature drop ($10 ext{--}14^\circ ext{C}$ below room return air).

**SAFETY WARNING:** Never top-up R-410A or R-32 without checking system integrity. Always wear protective gloves and safety goggles when handling high-pressure refrigerants to prevent frostbite and chemical exposure.

## 4. Weekly Activity Summary & Equipment Breakdown

| SERVICE CATEGORY | UNITS SERVICED | REFRIGERANT USED | AVG. COMPLETION TIME | STATUS |
|---|---|---|---|---|
| Routine PM & Deep Wash | 84 Units | N/A | 45 Mins | COMPLETED |
| Gas Leak Repair & R-410A Recharge | 24 Units | 19.2 kg R-410A | 90 Mins | COMPLETED |
| Gas Top-Up R-32 Systems | 14 Units | 8.4 kg R-32 | 60 Mins | COMPLETED |
| Capacitor & Fan Motor Replacement | 12 Units | N/A | 40 Mins | COMPLETED |
| Compressor Overhaul / Replacement | 8 Units | 7.2 kg R-410A | 180 Mins | IN PROGRESS |

## 5. Key Insights & Operational Recommendations

- **Flare Joint Quality:** Over 60% of gas leak calls were attributed to cracked flare nuts or improper copper flaring performed during initial installations. Re-flaring standard operating procedures have been reinforced across all field crews.
- **Digital Scale Usage:** Mandatory use of digital scales during liquid gas charging has reduced overcharging instances by 100%, lowering compressor strain and power consumption.
- **Inventory Request:** Recommend ordering 10 additional R-410A cylinders and replacement high-pressure gauge manifolds for Team B prior to next week.

**Report Compiled By:**

HVAC Field Supervisor

Date: August 13, 2026

**Approved By:**

Head of Residential Maintenance

Date: August 13, 2026